

Week - 1

Program:

```
1 • CREATE DATABASE IF NOT EXISTS week1;
2 • USE week1;
3
```

Output:

```
1 11:55:52 CREATE DATABASE IF NOT EXISTS week1 1 row(s) affected
2 11:56:00 USE week1 0 row(s) affected
```

Program:

```
• CREATE TABLE books(
    book_id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(255) NOT NULL,
    isbn VARCHAR(55) NOT NULL UNIQUE,
    published_year INT,
    CONSTRAINT chk_published_year CHECK (published_year < 2027)
);

• CREATE TABLE members(
    member_id INT AUTO_INCREMENT PRIMARY KEY,
    full_name VARCHAR(100) NOT NULL,
    email VARCHAR(160) NOT NULL UNIQUE
);
```

Output:

```
3 11:56:23 CREATE TABLE books( book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, isbn VARCHAR(5... 0 row(s) affected
4 11:56:31 CREATE TABLE members( member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, email VA... 0 row(s) affected
```

Program:

```
DESCRIBE books;
```

Output:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(255)	NO		NULL	
isbn	varchar(55)	NO	UNI	NULL	
published_year	int	YES		NULL	

```
8 12:01:11 DESCRIBE books
```

Program:

```
DESCRIBE members;
```

Output:

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	member_id	int	NO	PRI	NULL	auto_increment
	full_name	varchar(100)	NO		NULL	
	email	varchar(160)	NO	UNI	NULL	

9 12:02:37 DESCRIBE members

Program:

```
INSERT INTO books(title,isbn,published_year) VALUES  
( 'The Alchemist' , '9780061122415',1988),  
( 'Clean Code' , '9780132350884',2008),  
( 'Atomic Habits' , '97807354119',2018);
```

Output:

7 11:56:42 INSERT INTO books(title,isbn,published_year) VALUES ('The Alchemist' , '9780061122415',1988), ('Clean Code' , '9780132350884',2008), ('Atomic Habits' , '97807354119',2018);

Program:

```
INSERT INTO members (full_name, email) VALUES  
( 'Teju Roy' , 'teju@gmail.com'),  
( 'Sahasra' , 'sahasra@gmail.com'),  
( 'Pavani P' , 'pavani@gmail.com');
```

Output:

11 12:04:28 INSERT INTO members (full_name, email) VALUES ('Teju Roy' , 'teju@gmail.com'), ('Sahasra' , 'sahasra@gmail.com'), ('Pavani P' , 'pavani@gmail.com');

Program:

```
SELECT * FROM books;
```

Output:

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	97807354119	2018
NULL	NULL	NULL	NULL

✓ 12 12:05:07 SELECT * FROM books LIMIT 0, 1000

Program:

```
SELECT * FROM members;
```

Output:

	member_id	full_name	email
▶	1	Teju Roy	teju@gmail.com
	2	Sahasra	sahasra@gmail.com
	3	Pavani P	pavani@gmail.com
✱	NULL	NULL	NULL

✓ 13 12:06:02 SELECT * FROM members LIMIT 0, 1000

Program:

```
CREATE TABLE student(  
  student_id INT AUTO_INCREMENT PRIMARY KEY,  
  student_name VARCHAR(100) NOT NULL,  
  student_mobile VARCHAR(15) UNIQUE  
);  
  
INSERT INTO student(student_name,student_mobile) VALUES  
( 'Sahasra', '9898987722'),  
( 'Tejaswini', '8989892345'),  
( 'Pavani', '8899102345');
```

Output:

✓ 15 12:07:48 CREATE TABLE student(student_id INT AUTO_INCREMENT PRIMARY KEY, student_name VARCHAR(100) NOT NULL, student.
✓ 16 12:08:33 INSERT INTO student(student_name,student_mobile) VALUES ('Sahasra','9898987722'), ('Tejaswini','8989892345'), ('Pavani', '8899.

Program:

```
SELECT * FROM student;
```

Output:

	student_id	student_name	student_mobile
▶	1	Sahasra	9898987722
	2	Tejaswini	8989892345
	3	Pavani	8899102345

✓ 17 12:08:54 SELECT * FROM student LIMIT 0, 1000

Student Marks:

Program:

```
CREATE DATABASE IF NOT EXISTS week1_stdmarks;  
USE week1_stdmarks;  
CREATE TABLE student_marks (  
    roll_no INT PRIMARY KEY,  
    name VARCHAR(50),  
    subject VARCHAR(50),  
    marks DECIMAL(5,2)  
);
```

Output:

```
✓ 25 12:26:27 CREATE DATABASE IF NOT EXISTS week1_stdmarks  
✓ 26 12:26:42 USE week1_stdmarks  
✓ 27 12:28:46 CREATE TABLE student_marks ( roll_no INT PRIMARY KEY, name VARCHAR(50), subject VARCHAR(50),
```

Program:

```
INSERT INTO student_marks (roll_no, name, subject, marks) VALUES  
(1, 'Ravi', 'Math', 85.50),  
(2, 'Sita', 'Math', 92.75),  
(3, 'Anil', 'Math', 78.40),  
(4, 'Priya', 'Math', 88.90),  
(5, 'Vijay', 'Math', 80.25),  
(6, 'subbusir', 'aws', 98.50),  
(7, 'dwaraka', 'dbms', 95.75),  
(8, 'ranjani', 'english', 97.40),  
(9, 'kavithamam', 'aws1', 99.90),  
(10, 'seetha', 'azure', 82.25);
```

Output:

```
✓ 28 12:31:17 INSERT INTO student_marks (roll_no, name, subject, marks) VALUES (1, 'Ravi', 'Math', 85.50), (2, 'Sita', 'Math', 92.75), (3, 'Anil', 'Ma...
```

Program:

```
SELECT * FROM student_marks;
```

Output:

	roll_no	name	subject	marks
▶	1	Ravi	Math	85.50
	2	Sita	Math	92.75
	3	Anil	Math	78.40
	4	Priya	Math	88.90
	5	Vijay	Math	80.25
	6	subbusir	aws	98.50
	7	dwaraka	dbms	95.75
	8	ranjani	english	97.40
	9	kavithamam	aws1	99.90
	10	seetha	azure	82.25
*	NULL	NULL	NULL	NULL

✓ 29 12:31:43 SELECT * FROM student_marks LIMIT 0, 1000

Program:

```
SELECT COUNT(*) AS total_students FROM student_marks;
```

Output:

	total_students
▶	10

Program:

```
SELECT SUM(marks) FROM student_marks;
```

Output:

	SUM(marks)
▶	899.60

Program:

```
SELECT AVG(marks) FROM student_marks;
```

Output:

	AVG(marks)
▶	89.960000

Program:

```
SELECT MAX(marks) FROM student_marks;
```

Output:

	MAX(marks)
▶	99.90

Program:

```
SELECT MIN(marks) FROM student_marks;
```

Output:

	MIN(marks)
▶	78.40

Program:

```
SELECT * FROM student_marks WHERE marks > 85;
```

Output:

	roll_no	name	subject	marks
▶	1	Ravi	Math	85.50
	2	Sita	Math	92.75
	4	Priya	Math	88.90
	6	subbusir	aws	98.50
	7	dwaraka	dbms	95.75
	8	ranjani	english	97.40
	9	kavithamam	aws1	99.90
*	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM student_marks WHERE marks >= 90;
```

Output:

	roll_no	name	subject	marks
▶	2	Sita	Math	92.75
	6	subbusir	aws	98.50
	7	dwaraka	dbms	95.75
	8	ranjani	english	97.40
	9	kavithamam	aws1	99.90
*	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM student_marks WHERE marks < 80;
```

Output:

	roll_no	name	subject	marks
▶	3	Anil	Math	78.40
*	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;
```

Output:

	roll_no	name	subject	marks
▶	1	Ravi	Math	85.50
	4	Priya	Math	88.90
	5	Vijay	Math	80.25
	10	seetha	azure	82.25
•	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM student_marks WHERE name LIKE 'P%';
```

Output:

	roll_no	name	subject	marks
▶	4	Priya	Math	88.90
•	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay');
```

Output:

	roll_no	name	subject	marks
▶	1	Ravi	Math	85.50
	2	Sita	Math	92.75
	5	Vijay	Math	80.25
•	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM student_marks WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');
```

Output:

	roll_no	name	subject	marks
▶	1	Ravi	Math	85.50
	2	Sita	Math	92.75
	4	Priya	Math	88.90
•	NULL	NULL	NULL	NULL

Program:

```
UPDATE student_marks SET marks = 90.00 WHERE roll_no = 3;  
UPDATE student_marks SET subject = 'Remedial Math' WHERE marks < 80;
```

Output:

```
✓ 42 12:40:47 UPDATE student_marks SET marks = 90.00 WHERE roll_no = 3  
✓ 43 12:40:51 UPDATE student_marks SET subject = 'Remedial Math' WHERE marks < 80
```

Program:

```
DELETE FROM student_marks WHERE roll_no = 5;  
DELETE FROM student_marks WHERE marks < 75;
```

Output:

```
✓ 44 12:41:45 DELETE FROM student_marks WHERE roll_no = 5  
✓ 45 12:41:47 DELETE FROM student_marks WHERE marks < 75
```

Program:

```
SELECT * FROM student_marks ORDER BY marks ASC;
```

Output:

	roll_no	name	subject	marks
▶	10	seetha	azure	82.25
	1	Ravi	Math	85.50
	4	Priya	Math	88.90
	3	Anil	Math	90.00
	2	Sita	Math	92.75
	7	dwaraka	dbms	95.75
	8	ranjani	english	97.40
	6	subbusir	aws	98.50
	9	kavithamam	aws1	99.90
✱	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM student_marks ORDER BY marks DESC;
```

Output:

	roll_no	name	subject	marks
▶	10	seetha	azure	82.25
	1	Ravi	Math	85.50
	4	Priya	Math	88.90
	3	Anil	Math	90.00
	2	Sita	Math	92.75
	7	dwaraka	dbms	95.75
	8	ranjani	english	97.40
	6	subbusir	aws	98.50
	9	kavithamam	aws1	99.90
*	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM student_marks ORDER BY marks DESC;
```

Output:

	roll_no	name	subject	marks
▶	9	kavithamam	aws1	99.90
	6	subbusir	aws	98.50
	8	ranjani	english	97.40
	7	dwaraka	dbms	95.75
	2	Sita	Math	92.75
	3	Anil	Math	90.00
	4	Priya	Math	88.90
	1	Ravi	Math	85.50
	10	seetha	azure	82.25
*	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM student_marks ORDER BY name ASC;
```

Output:

	roll_no	name	subject	marks
▶	3	Anil	Math	90.00
	7	dwaraka	dbms	95.75
	9	kavithamam	aws1	99.90
	4	Priya	Math	88.90
	8	ranjani	english	97.40
	1	Ravi	Math	85.50
	10	seetha	azure	82.25
	2	Sita	Math	92.75
	6	subbusir	aws	98.50
•	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM student_marks ORDER BY name DESC;
```

Output:

	roll_no	name	subject	marks
▶	6	subbusir	aws	98.50
	2	Sita	Math	92.75
	10	seetha	azure	82.25
	1	Ravi	Math	85.50
	8	ranjani	english	97.40
	4	Priya	Math	88.90
	9	kavithamam	aws1	99.90
	7	dwaraka	dbms	95.75
	3	Anil	Math	90.00
•	NULL	NULL	NULL	NULL

Program:

```
SELECT subject, SUM(marks) AS total_marks FROM student_marks GROUP BY subject;
```

Output:

	subject	total_marks
▶	Math	357.15
	aws	98.50
	dbms	95.75
	english	97.40
	aws1	99.90
	azure	82.25

Program:

```
SELECT subject, AVG(marks) AS avg_marks
FROM student_marks
GROUP BY subject
HAVING AVG(marks) > 90;
```

Output:

	subject	avg_marks
▶	aws	98.500000
	dbms	95.750000
	english	97.400000
	aws1	99.900000

Program:

```
SELECT subject, COUNT(*) AS num_students
FROM student_marks
GROUP BY subject
HAVING COUNT(*) > 1;
```

Output:

	subject	num_students
▶	Math	4